

RITICA AWASTHI

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Professional Summary

Final-year B.Tech Computer Science (AI & ML) student who enjoys working with data and building practical machine learning solutions. Experienced in supervised learning, model evaluation, and deployment, with hands-on exposure during an internship at Hindustan Copper Limited.

Technical Skills

- Programming Languages:** Python, C++, Java, SQL
- Machine Learning:** PyTorch, TensorFlow, XGBoost, Scikit-learn, Pandas, NumPy, SHAP, Isolation Forest, Model Evaluation, Feature Engineering, Model Deployment, Computer Vision
- Generative AI:** GPT-4o, Groq/Llama 3, Prompt Engineering, RAG
- Backend & APIs:** FastAPI, REST APIs, SQLAlchemy
- Frontend:** React.js, TypeScript, Tailwind CSS, Vite, Streamlit
- Cloud & DevOps:** Git, AWS, GCP, Vercel, Render, Docker
- Tools:** Excel, GitHub, VS Code, Jupyter Notebook
- Soft Skills:** Team Collaboration, Communication, Problem-Solving, Critical Thinking, Leadership

Work Experience

Hindustan Copper Limited May 2026 – June 2026
AI/ML Intern (Group Project) — RecovAI: AI Decision Support System

- Developed and deployed an end-to-end production ML platform for copper recovery forecasting, reagent dosing optimization, and real-time anomaly detection across 26 process variables — achieving $R^2 = 0.972$ on 1,778 shift logs; approved for plant-wide production deployment.
- Architected a 5-engine ML pipeline integrating XGBoost, SHAP-based Explainable AI, Isolation Forest, and PSI drift monitoring, reducing false anomaly alerts by ~40% vs. threshold-based detection.
- Designed and shipped 15+ FastAPI endpoints with SQLAlchemy persistence powering a live Chart.js dashboard, cutting per-shift engineer reporting time by ~60%.
- Automated shift-report generation using an LLM-powered workflow on Groq/Llama 3 with structured prompt engineering — reducing manual documentation effort by ~70% for plant staff.

Projects

Tech Jargon Translator — CS Learning Hub GitHub
React, TypeScript, Tailwind CSS, Vite, shadcn/ui, Chart.js

- Built a search-first learning hub for CS students that combines instant jargon lookup with structured topic browsing across 7 sections (OS, Networks, DBMS, Cloud, AWS, Programming Languages, AI & LLMs).
- Curated and visualized 90+ beginner-friendly terms with real diagrams, difficulty badges, related-term recommendations, and responsive detail pages.
- Implemented fuzzy site-wide search with autocomplete, subtopic filtering, and breadcrumb navigation; optimized for fast lookup and structured learning paths.

Pixora — AI Image Enhancement Suite GitHub
React, FastAPI, OpenCV, PyTorch, Real-ESRGAN, GFPGAN, Rembg, Diffusers

- Built and deployed a full-stack AI image-enhancement platform exposing REST APIs for super-resolution (4× upscaling), background removal, HDR tone-mapping, and face restoration using state-of-the-art diffusion and GAN-based models.
- Unified multiple deep-learning pipelines behind a single FastAPI inference backend with a React drag-and-drop UI, real-time before/after comparison, and average processing time under 3 seconds; hosted on Vercel and Hugging Face Spaces.

Education

B.Tech — Computer Science (AI & ML) — VIT Bhopal University	CGPA: 8.51/10	2023 – present
Higher Secondary Certificate (HSC) — Kendriya Vidyalaya Aliganj, Lucknow		84.4%
Secondary School Certificate (SSC) — Kendriya Vidyalaya Aliganj, Lucknow		91.2%

Certifications

- Cloud Computing — NPTEL/IIT Kharagpur (2025)
- AWS Technical Essentials — Amazon Web Services (2025)
- Data Science in Python — University of Michigan/Coursera (2025)
- Google IT Support Professional Certificate — Google (2026)
- Computer Vision — Vityarthi (2026)